22 A uniform wire is made of a metal that has a Young modulus of 1.3×10^{11} Pa.

The wire is 2.4 m long and has a spring constant of 2.7×10^{4} N m^{-1}.

What is the volume of the wire?

A 2.6×10^{-8} m^3 **B** 6.3×10^{-8} m^3 **C** 5.0×10^{-7} m^3 **D** 1.2×10^{-6} m^3